

# ISSS602\_AY2025-26Aug\_Assign1\_ES

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# HappyRides

Done By: Nor Hendra Bin Abdul Rahman

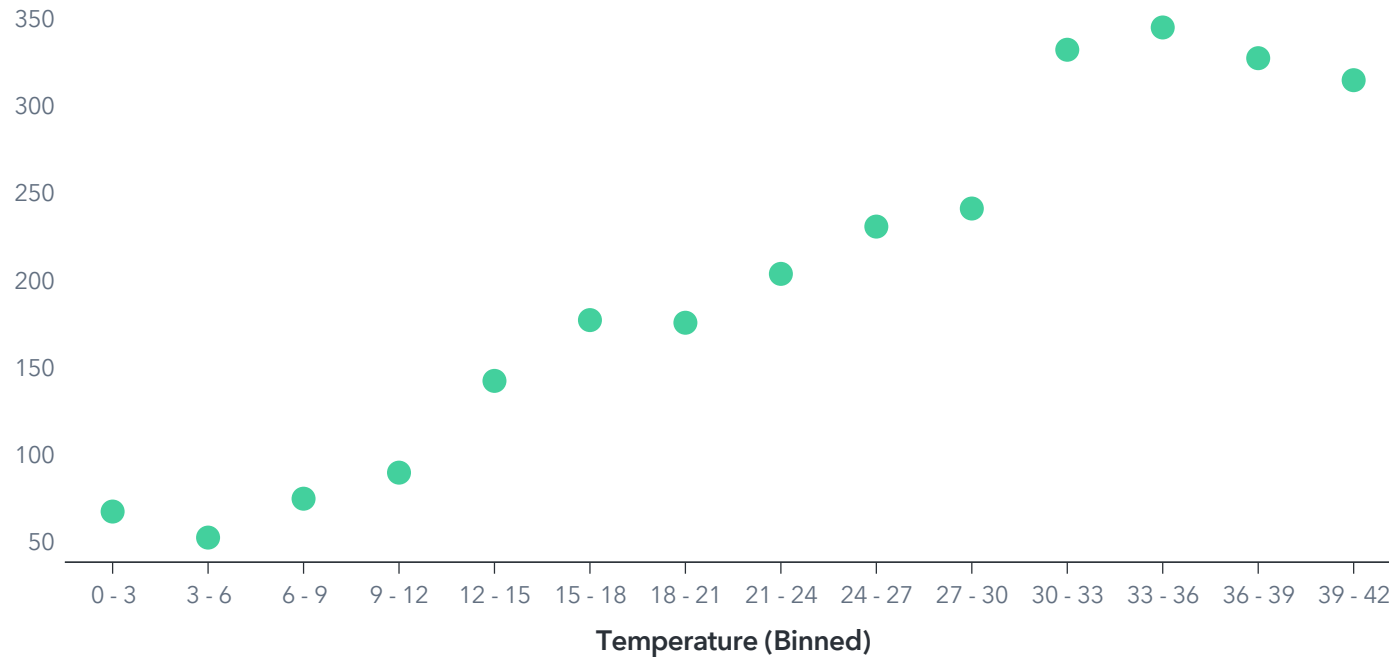
## Objective

This report investigates what drives hour-to-hour demand for HappyRides and translates the findings into operational guidance. The analysis examines meteorological conditions (weather class, humidity, temperature, wind), seasonality, day type, time-of-day patterns, and rider segments to understand how each factor shapes usage and when demand concentrates. Using confirmatory data analysis techniques (ANOVA,  $\chi^2$  tests, t-tests, correlation, and distributional profiling), the report identifies the most influential levers for forecasting and capacity planning, proposes simple KPIs that can be monitored in real time, and highlights targeted data improvements.

# Weather & Temperature Impact on Users

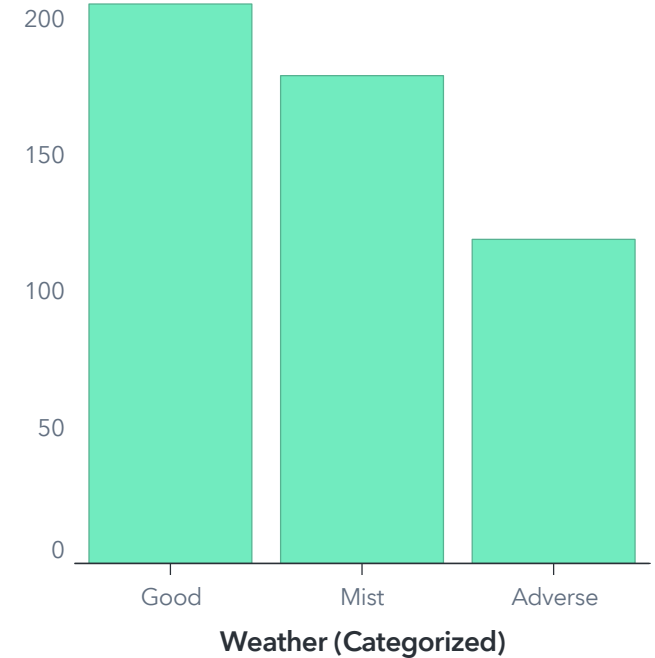
## Average Users by Temperature (Binned)

Average Users



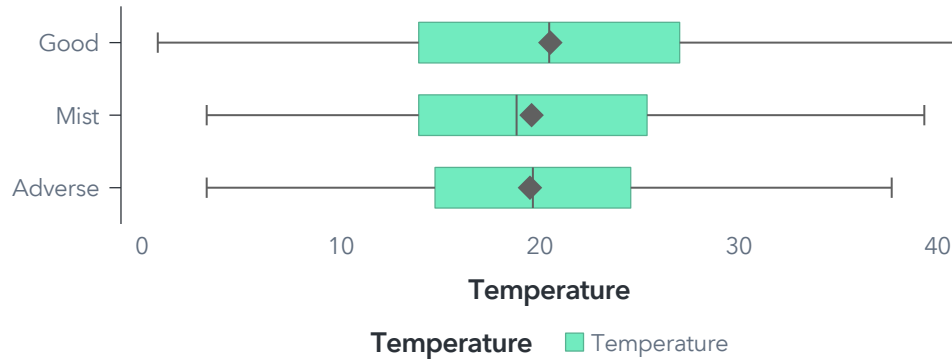
## Users by Weather (Categorized)

Average Users



## Temperature by Weather (Categorized)

Weather (Categorized)



### Insight 1:

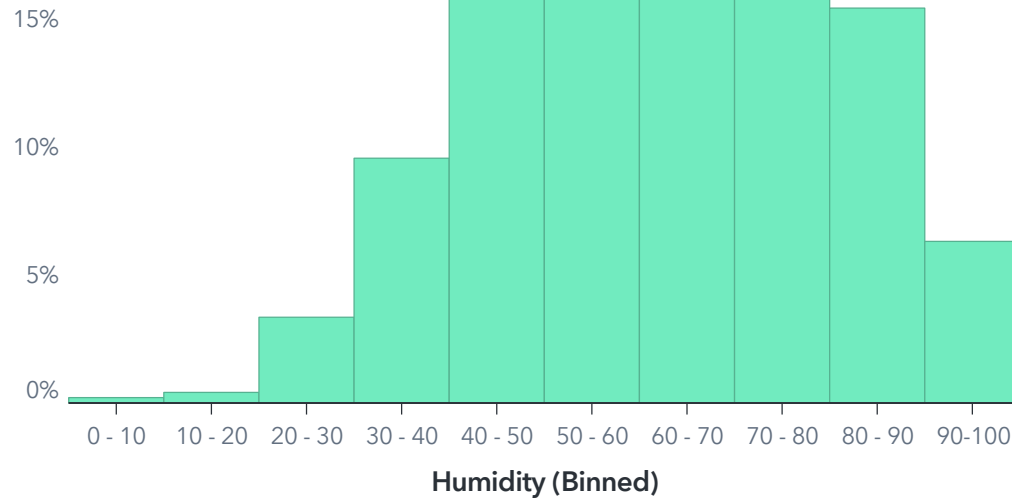
**Weather is a clear demand driver:** a Welch ANOVA shows  $F(2, 2452.5)=140.97$ ,  $p<.0001$ , with hourly means Good=205.24, Mist=178.96, Adverse=118.90; all pairwise differences are significant (Tukey-Kramer  $p<.0001$ ). Although the classical fit yields  $R^2\approx0.0177$ , the bar chart exhibits a consistent Good > Mist > Adverse ordering, indicating a strong directional effect even amid other contributing factors.

### Insight 2:

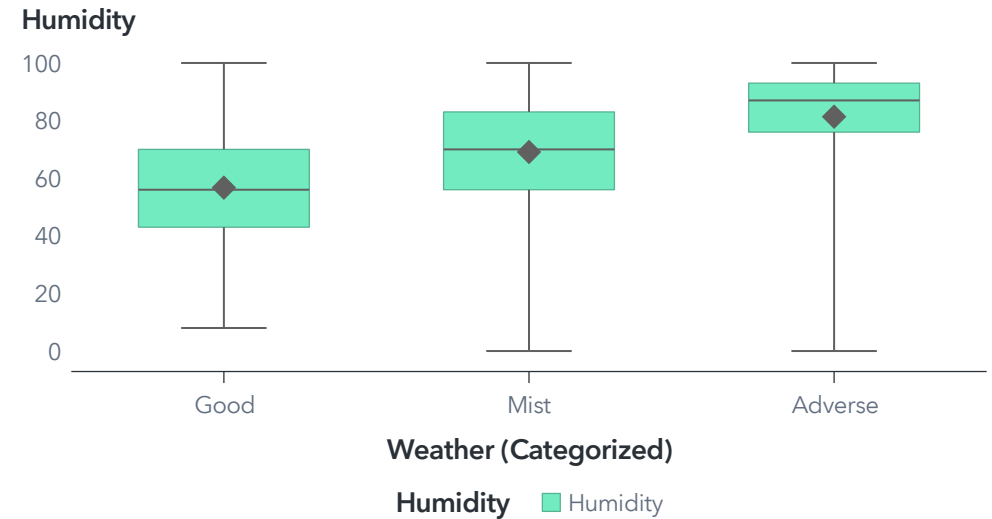
The boxplot shows distinct temperature distributions under each weather state (Good concentrated in mid-temperatures with tighter IQRs; Mist cooler and broader; Adverse sparser and more variable). The association is present but **weak** on the contingency test ( $\chi^2(26)=301.78$ ,  $p<.0001$ ; Cramér's  $V\approx0.118$ ). Read alongside the temperature curve, average users **increase with temperature and peak at 30-33 °C, then soften at 36-42 °C**, indicating a comfort band while the prevailing weather regime mainly shifts how often those temperatures occur.

# Humidity Exposure and Effect

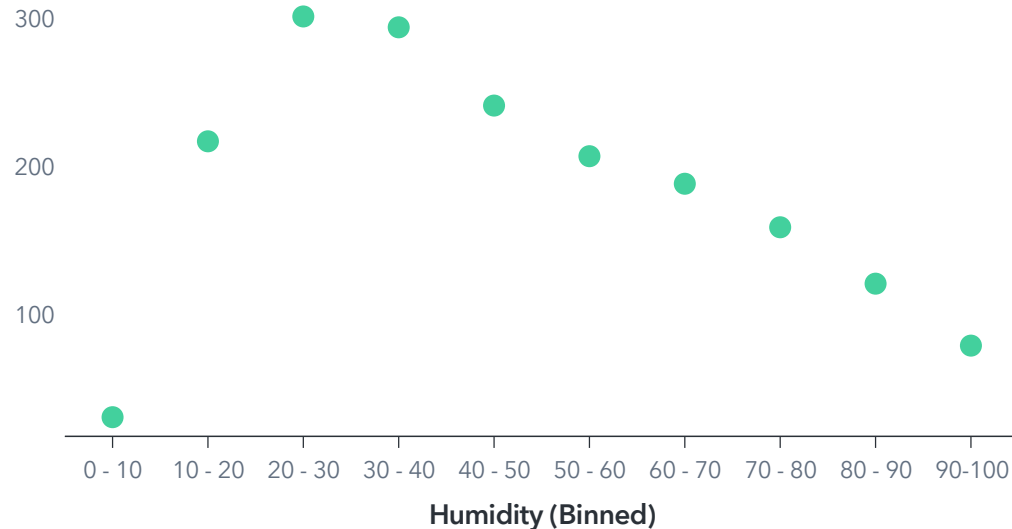
Hours by Humidity (%)  
% of Hours



Humidity by Weather (Categorized)



Average Users by Humidity  
Average Users



## Insight 3:

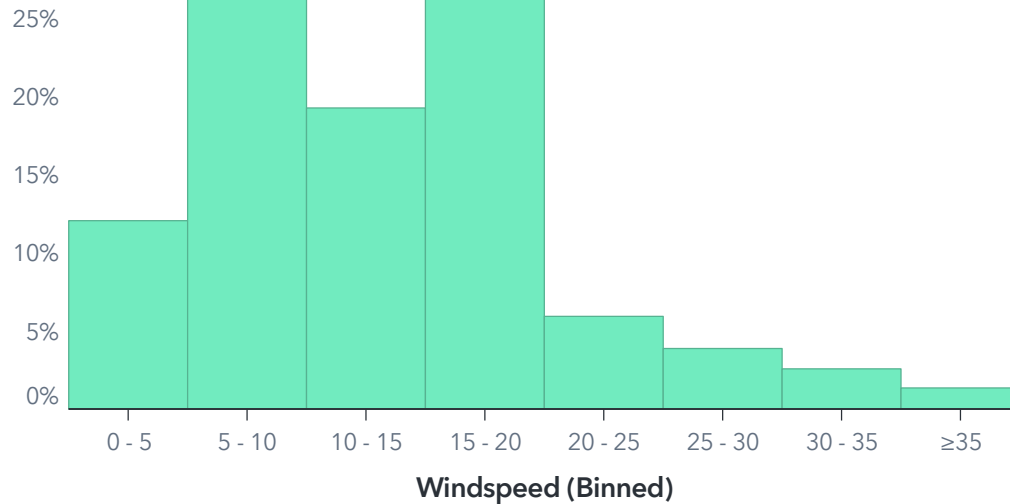
The histogram shows most hours fall between 40-80% RH. The boxplots indicate **distinct humidity profiles by weather**: Good clusters in mid-humidity with tighter IQRs, Mist is slightly higher and broader, and Adverse shifts markedly upward with frequent 80 - 100% RH. This composition difference is statistically strong in the contingency test of Humidity (Binned) × Weather (Categorized):  $\chi^2(18)=2709.37$ ,  $p<.0001$ , with **Cramér's V=0.353** indicating a **moderate association**.

## Insight 4:

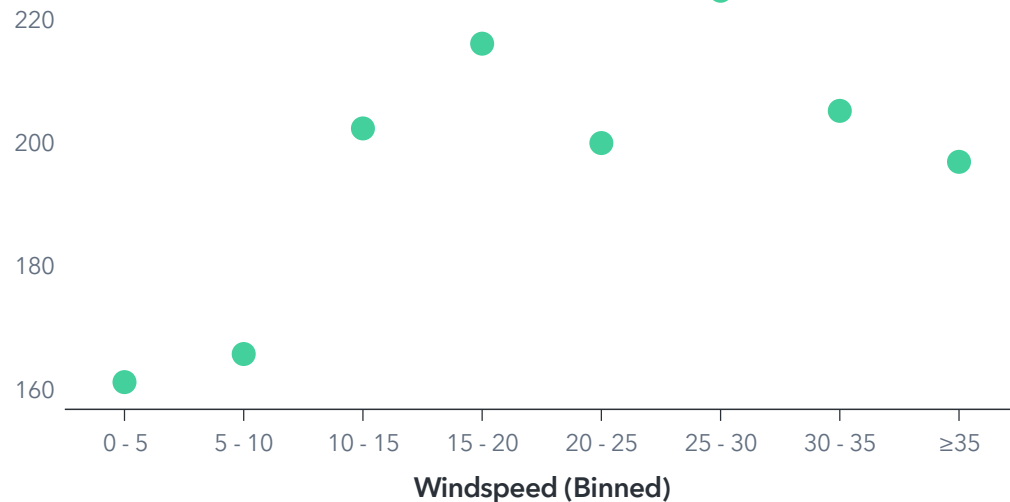
The **users-by-humidity plot rises to a peak in the 20 - 30% and 30 - 40% bins**, then declines steadily as humidity increases. Bin means quantify the shape: 20-30% = **301.5** users/hr ( $n=364$ ), 30-40% = **294.2** ( $n=1039$ ), 40-50% = **241.3** ( $n=1727$ ), 50-60% = **207.0** ( $n=1842$ ), 60-70% = **188.4** ( $n=1748$ ), 70-80% = **158.9** ( $n=1736$ ), 80-90% = **120.8** ( $n=1676$ ), 90-100% = **78.9** ( $n=686$ ). Differences across bins are significant on Welch's ANOVA,  $F(9, 478.6)=240.89$ ,  $p<.0001$ , with the classical ANOVA reporting  $F(9, 10876)=150.93$ ,  $p<.0001$  and  $R^2\approx 0.111$ . Variances are unequal (Levene  $p<.0001$ ), so the Welch result is the primary inference.

## Wind Exposure and Effect

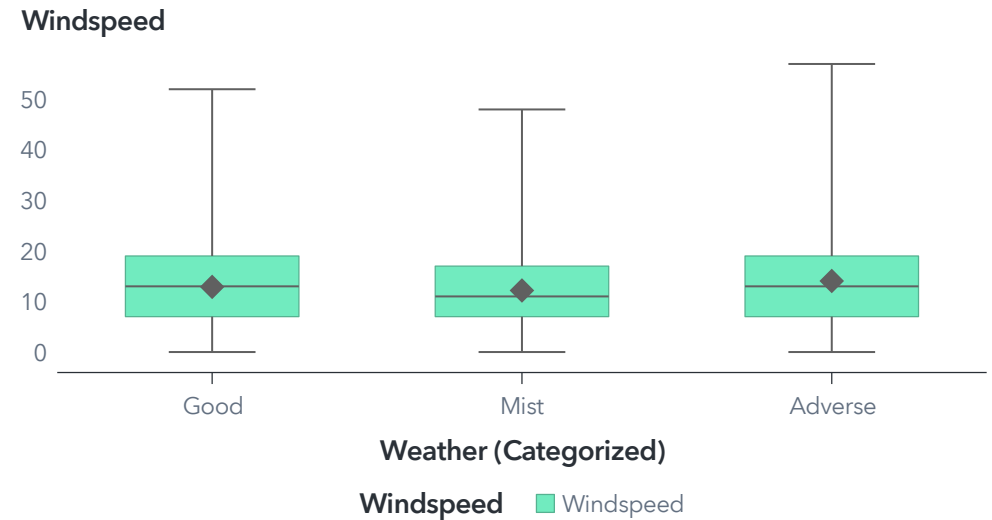
Hours by Windspeed  
% of Hours



Average Users by Windspeed  
Average Users



Windspeed by Weather (Categorized)



### Insight 5:

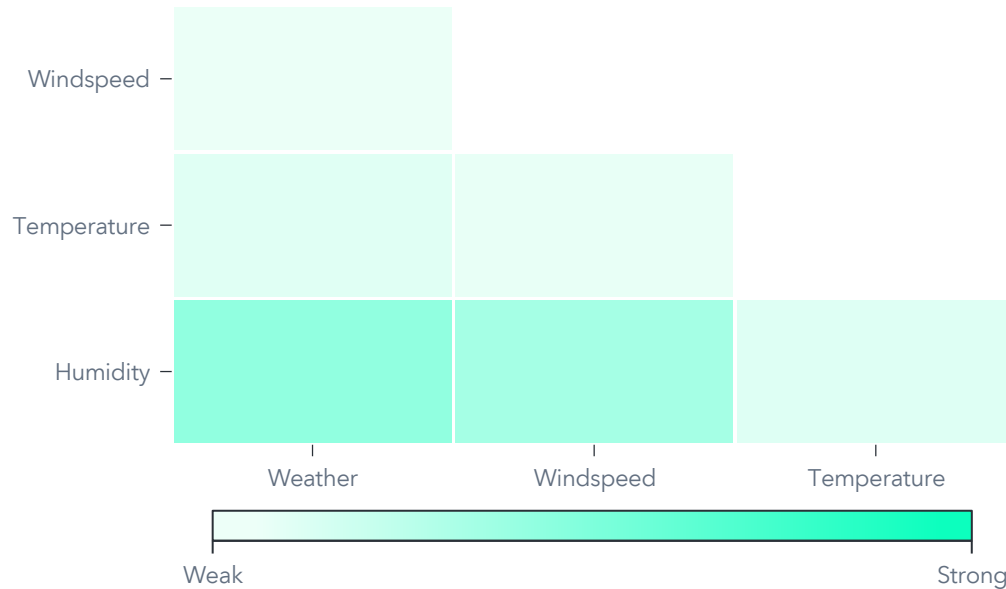
The histogram shows most hours occur between 5-20 units of windspeed, with fewer hours above 25. The boxplots indicate similar central tendencies across weather states, with Adverse displaying a slightly higher median and wider spread. This composition difference is **statistically detectable but weak** in the contingency test of Windspeed (binned) × Weather (categorized):  $\chi^2(14)=68.998$ ,  $p<.0001$ , Cramér's  $V=0.056$ .

### Insight 6:

**Per-hour usage rises from calm conditions to a mid-range plateau and then holds roughly steady:** means by bin are 0-5=161, 5-10=166, 10-15=202, 15-20=216, 20-25=200, 25-30=225, 30-35=205, ≥35=197 users/hr. **Differences are significant** on Welch's ANOVA,  $F(7, 1410.9)=26.23$ ,  $p<.0001$ , with the classical model reporting  $R^2\approx0.016$ . Tukey comparisons show the calm bins (0-5, 5-10) are lower than many mid-range bins, while several adjacent mid-range bins are not statistically distinct from each other, consistent with a broad "moderate wind" plateau rather than a sharp optimum.

# Meteorology Interplay

## Weather & Meteorology Association

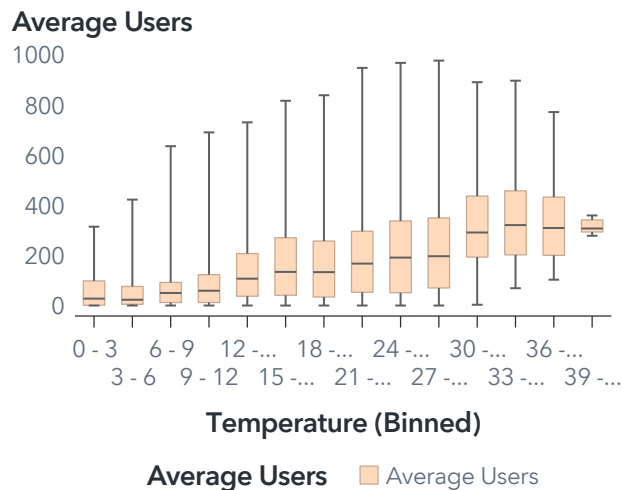


### Insight 7:

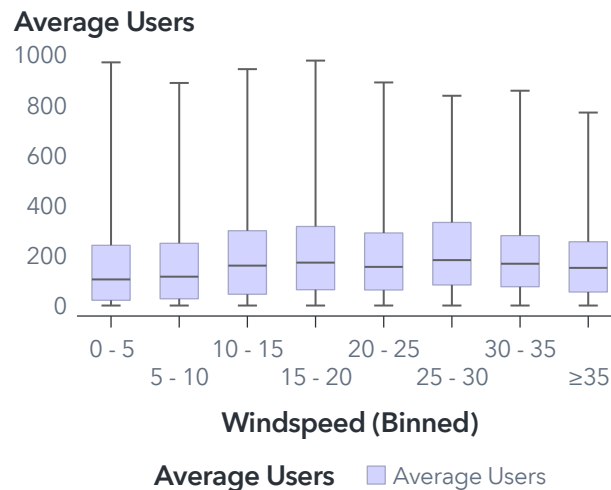
The mosaic plot is darker for Humidity by Weather and Humidity by Windspeed, which matches the stats: **humidity's composition varies most by weather** ( $\chi^2(18)=2709.37$ ,  $p<.0001$ ; Cramér's  $V=0.353$ , moderate), and **humidity and windspeed have a moderate negative relationship** (Pearson  $r = -0.319$ ; Spearman  $r = -0.324$ ; both  $p<.0001$ ). By contrast, **Temperature and Weather is weaker** ( $\chi^2(26)=301.78$ ,  $p<.0001$ ;  $V\approx0.118$ ) and **windspeed's shift by weather is very weak** ( $\chi^2(14)=68.998$ ,  $p<.0001$ ;  $V\approx0.056$ ). **Temperature correlates only slightly with humidity** (Pearson  $r = -0.065$ ) and is **near zero with windspeed** (Pearson  $r \approx -0.018$ ). Taken together, weather categories in this dataset mostly reflect the humidity regime, windspeed counter-moves with humidity, and temperature is comparatively orthogonal to both. This explains why humidity shows the clearest monotonic demand pattern, windspeed shows a broad mid-range plateau with limited direct linkage to weather, and temperature exhibits a comfort-band peak that is only lightly conditioned by weather.

The three boxplots summarize the effects seen earlier: usage climbs with temperature into the mid-20s °C before easing above about 27-30, declines almost monotonically as humidity rises with a marked drop beyond roughly 60-70, and is lower in very calm wind then broadly stable across a mid range of 10-30.

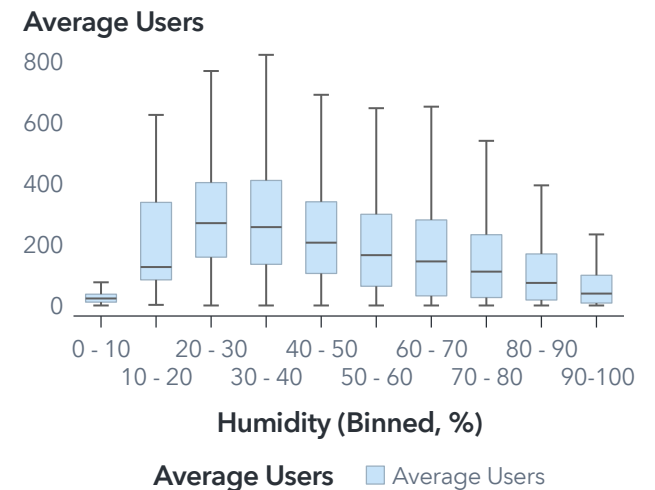
## Average Users by Temperature (Binned)



## Average Users by Windspeed (Binned)



## Average Users by Humidity (Binned)

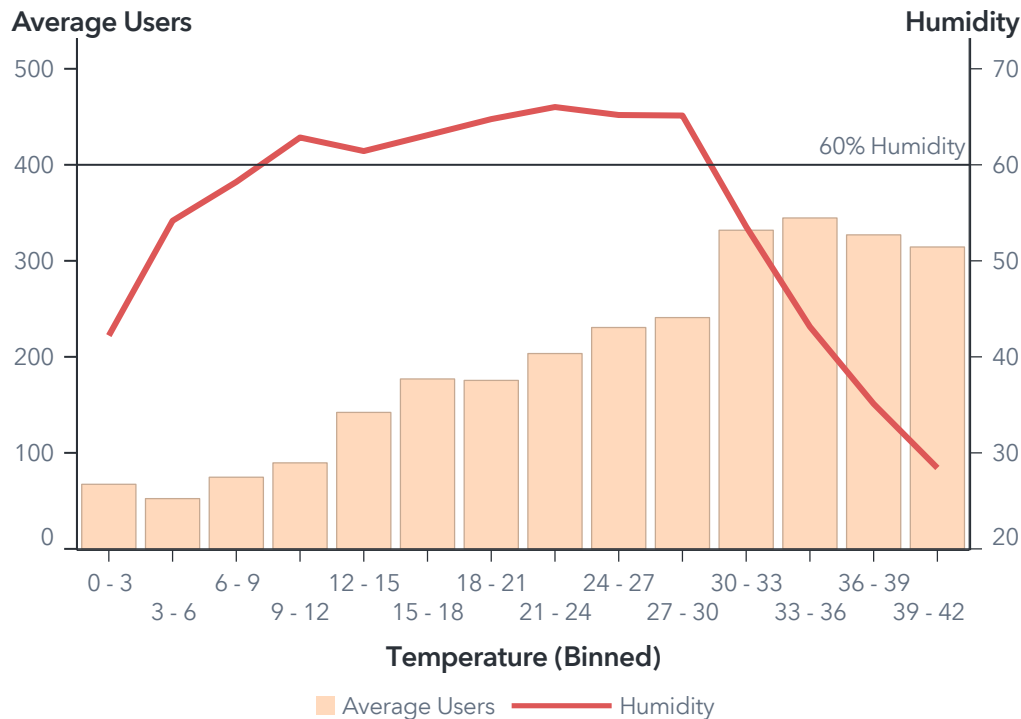


## Weather Stress Test: Heat, Humidity and Wind

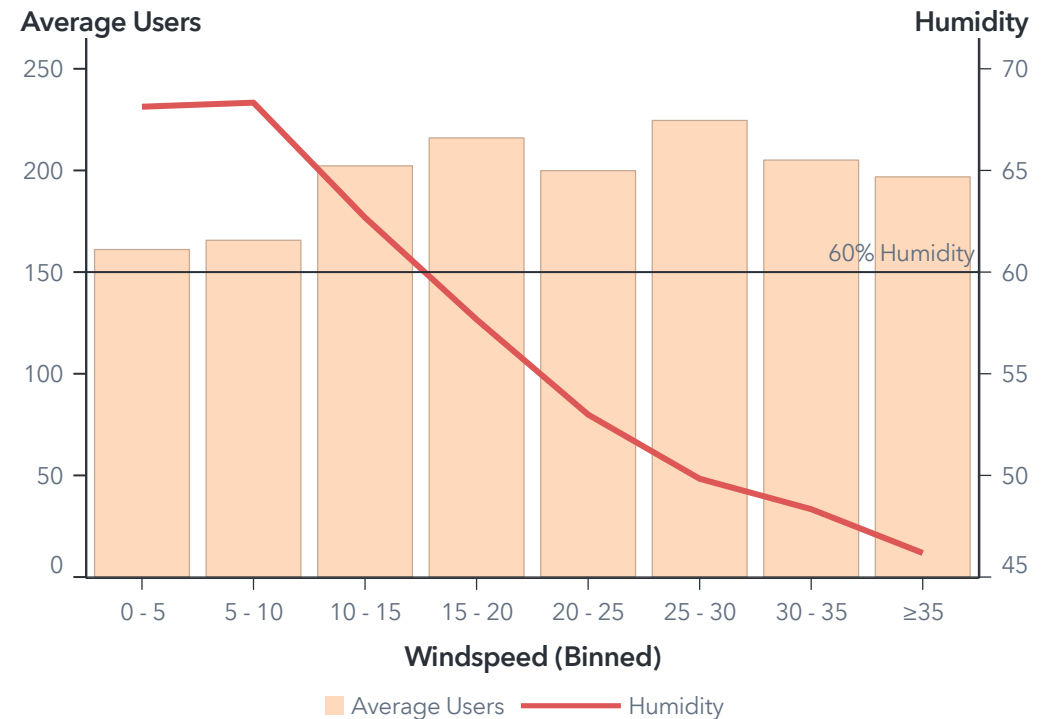
### Insight 8:

The **dual-axis bars and line show average users alongside mean humidity within each temperature and windspeed bin to test co-movement**. 60% relative humidity is used as the reference because prior research reports reduced bike-share demand once humidity exceeds about 60%, especially alongside higher temperatures. The split is also borne out here: hours at  $\leq 60\%$  average  $\approx 241$  users/hr while  $>60\%$  average  $\approx 146$  users/hr; the gap is  $\approx 95$  users/hr, significant on pooled and Satterthwaite t-tests ( $p < .0001$ ) and on Wilcoxon. A sweep of alternative cutoffs from 40% to the mid-70s produces similarly large and significant gaps, indicating that 60% is a reasonable, literature-grounded and data-supported pivot.

Average Users, Humidity by Temperature



Average Users, Humidity by Windspeed

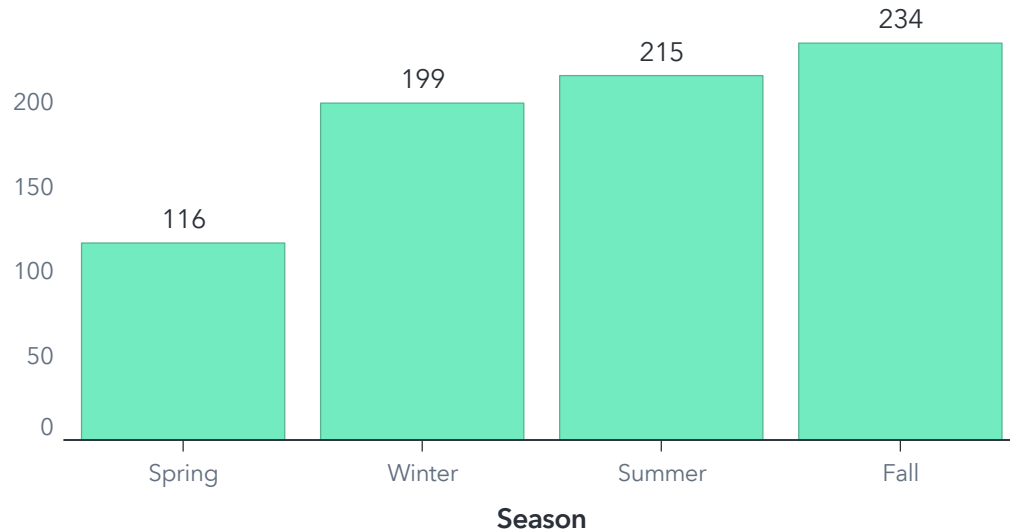


### Insight 9:

In the **temperature** panel, average users rise steadily and **peak at 30-33 °C**. Humidity climbs into the low-60s before that peak, then **falls below 60%** in the hottest bins; the modest softening in users at **36-42 °C** occurs while humidity is dropping, so the high-temperature effect is not explained by being above 60% RH in these bins. In the **windspeed** panel, the humidity line **decreases monotonically** with stronger winds and crosses below 60% around 15-20 units, yet average users form a **broad mid-range plateau** with a local high near **25-30**; demand does not mirror the falling humidity here. Taken together, humidity at 60% is a strong overall discriminator of demand, but within these stratifications the temperature peak and the wind plateau are only partially mediated by humidity.

# Seasonal Sweet Spot

Average Users by Season  
Average Users



## Insight 10:

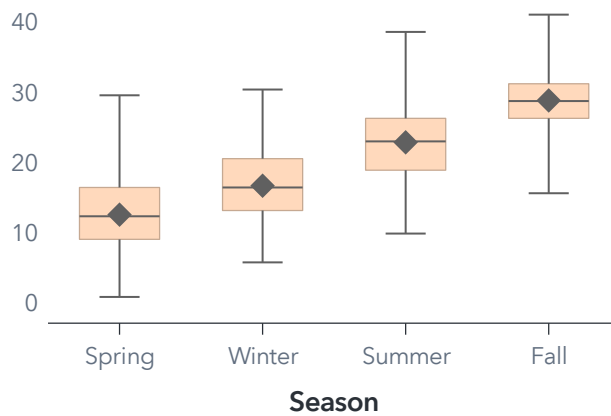
Average users vary materially by season: **Fall = 234, Summer = 215, Winter = 199, Spring = 116** users/hr. Welch's ANOVA confirms a season effect ( $F \approx 335.09$ ,  $p < .0001$ ), with all pairwise differences significant (Tukey-Kramer  $p < .001$ ). The classical fit shows  $R^2 \approx 0.061$ , so the **season signal is clear but explains a modest share of variance, consistent with other drivers at play**. The bar chart aligns with these means: Fall highest, Spring lowest.

## Insight 11:

Temperature varies strongly by season (Welch  $F \approx 6874.04$ ,  $p < .0001$ ): **Fall 28.79°C, Summer 22.82°C, Winter 16.65°C, Spring 12.53°C**, matching the higher demand in warmer seasons. Humidity also differs (Welch  $F \approx 137.34$ ,  $p < .0001$ ): **Winter 66.17%, Fall 64.12%, Summer 60.85%, Spring 56.30%**, though the effect size is small ( $R^2 \approx 0.037$ ). Windspeed shifts are statistically detectable but small in magnitude (Welch  $F \approx 87.35$ ,  $p < .0001$ ;  $R^2 \approx 0.025$ ), with **Spring** the windiest (14.64) and **Fall/Winter** the calmest (~11.5–11.7). Taken together, **the seasonal demand ordering lines up most closely with the temperature profile, with humidity and windspeed playing secondary roles**.

Temperature by Season

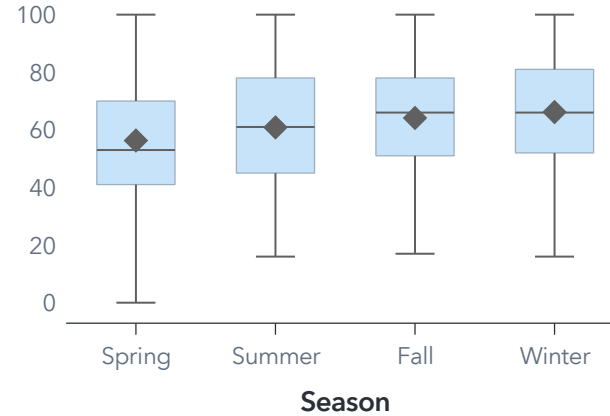
Temperature



Temperature Temperature

Humidity by Season

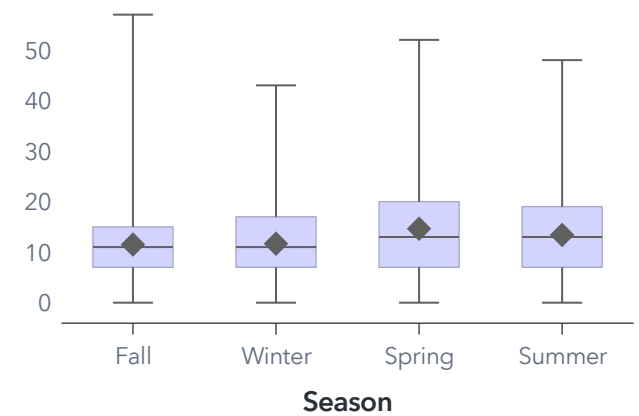
Humidity



Humidity Humidity

Windspeed by Season

Windspeed



Windspeed Windspeed

## Commute Peaks & Weekend Flow

Peak Expected User

**977**

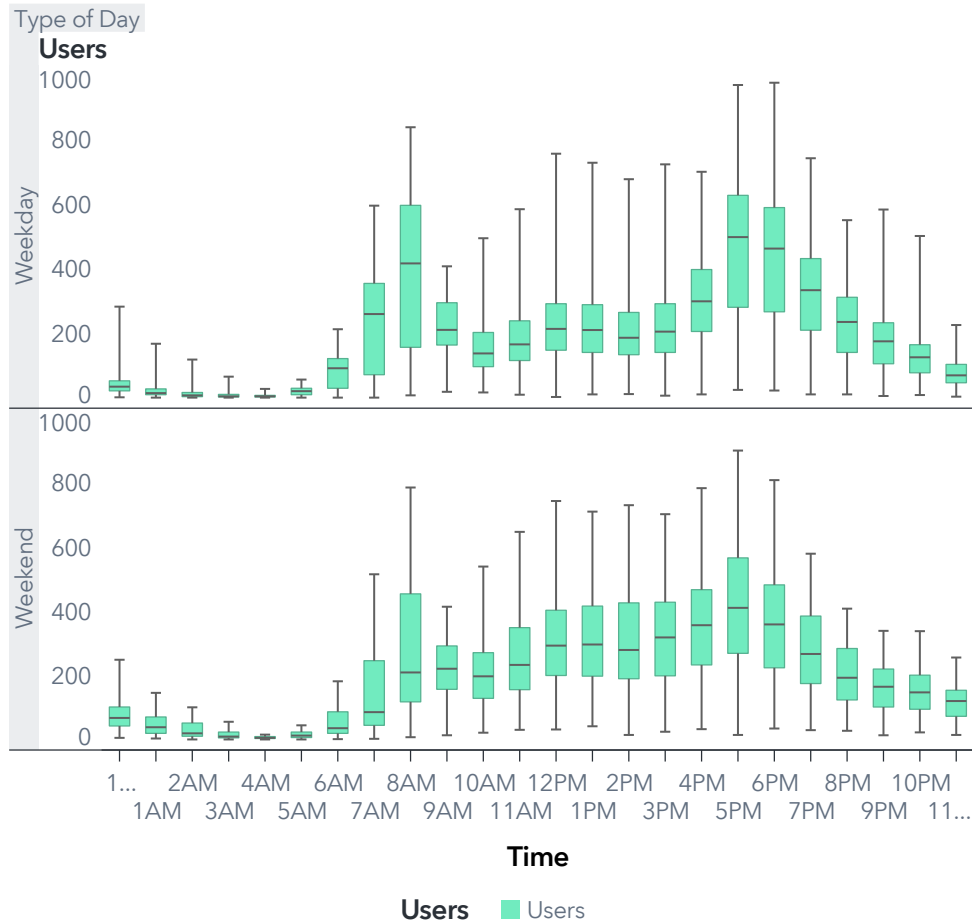
Type of Day: Weekday

Peak Expected User

**900**

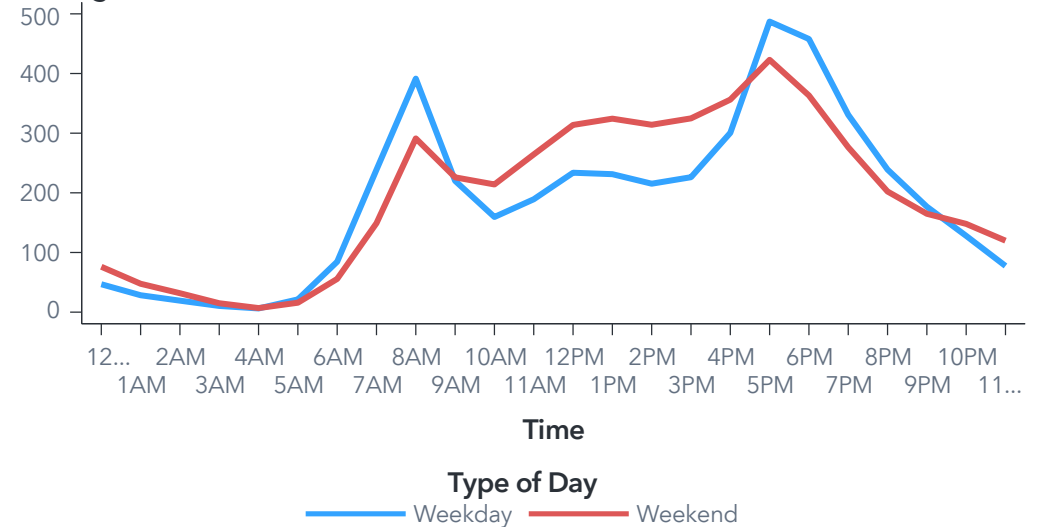
Type of Day: Weekend

### Average Users by Hour – Distribution



### Average Users by Time – Weekday vs Weekend

Average Users



#### Insight 12:

Weekends average **197.2** users/hr and weekdays **189.3** users/hr, a small but statistically detectable weekend lead (pooled  $t \approx -2.07$ ,  $p \approx 0.0387$ ; Satterthwaite  $t \approx -2.10$ ,  $p \approx 0.0360$ ; Wilcoxon  $Z = 3.52$ ,  $p = 0.0002$ ). The Peak Expected User KPI answers a different question than the mean: it indicates the single-hour load to provision for. Peaks are **977** on weekdays and **900** on weekends, showing **higher topside pressure on workdays** even though the pooled mean is slightly higher on weekends.

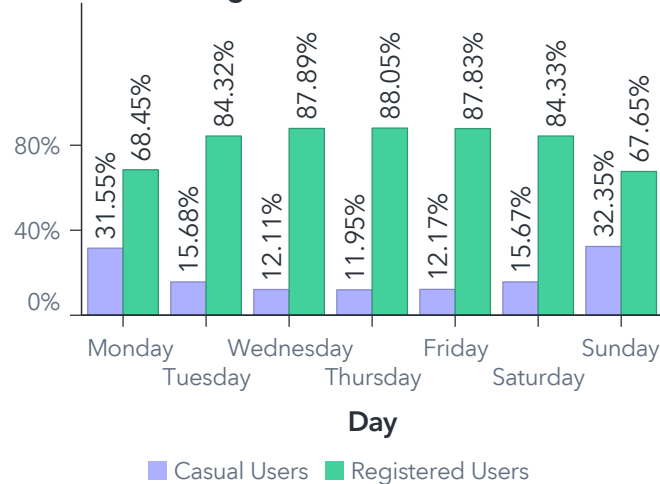
#### Insight 13:

After pairing by hour (24 matched slots), the mean difference **weekday – weekend = -8.51** users/hr is **not significant** ( $t(23) = -0.69$ ,  $p = 0.499$ ). The shape explains this result: **weekdays concentrate demand into sharp commute spikes in the morning and evening, while weekends lift later and sustain a broader mid-afternoon to evening plateau**. These crossing patterns offset in the hour-matched comparison, yielding no overall advantage by day type despite the higher weekday peak.

## Who Rides When – Registered vs Casual

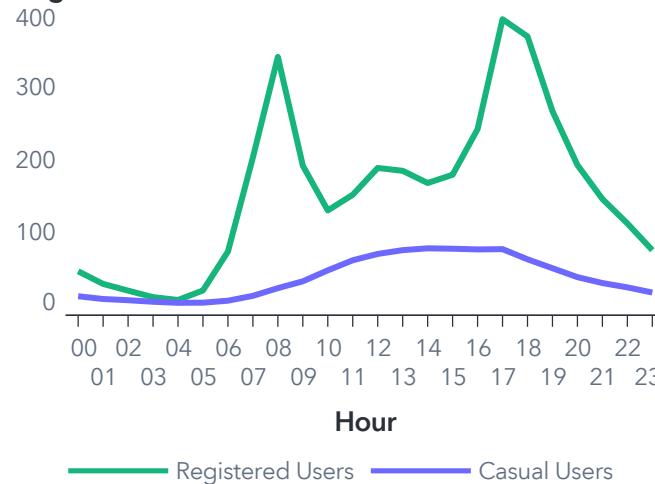
### Average Users by Day – Registered vs Casual

#### Casual Users / Registered Users



### Average Registered Users, Casual Users by Hour

#### Registered Users / Casual Users

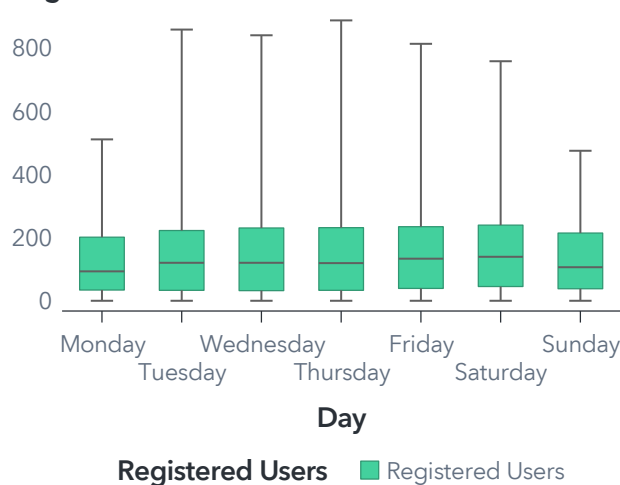


#### Insight 14:

Registered riders dominate every day of the week and drive the sharp commute shape, with pronounced AM and PM peaks in the hourly lines, while casual riders rise later and concentrate in the mid-afternoon to early evening. The stacked bars show registered share above roughly four-fifths on weekdays and lower on weekends, where casual riders expand to about one-third by Sunday. Registered users average **158** on weekdays vs **150** on weekends, while casual users average **31** on weekdays vs **48** on weekends, indicating a weekend shift toward casual usage rather than a reversal of dominance.

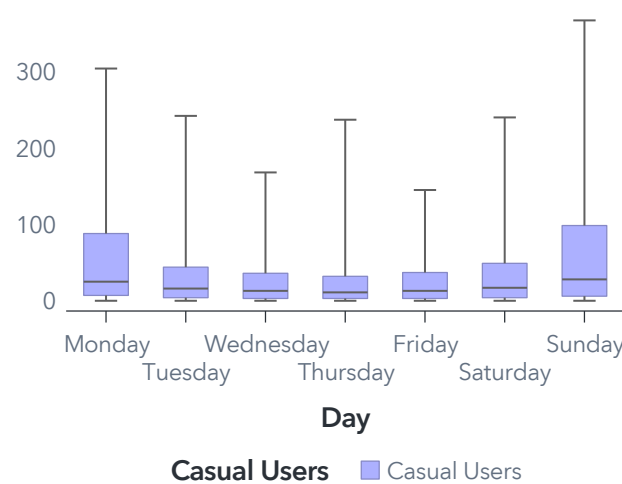
### Average Registered Users by Day

#### Registered Users



### Average Casual Users by Day

#### Casual Users

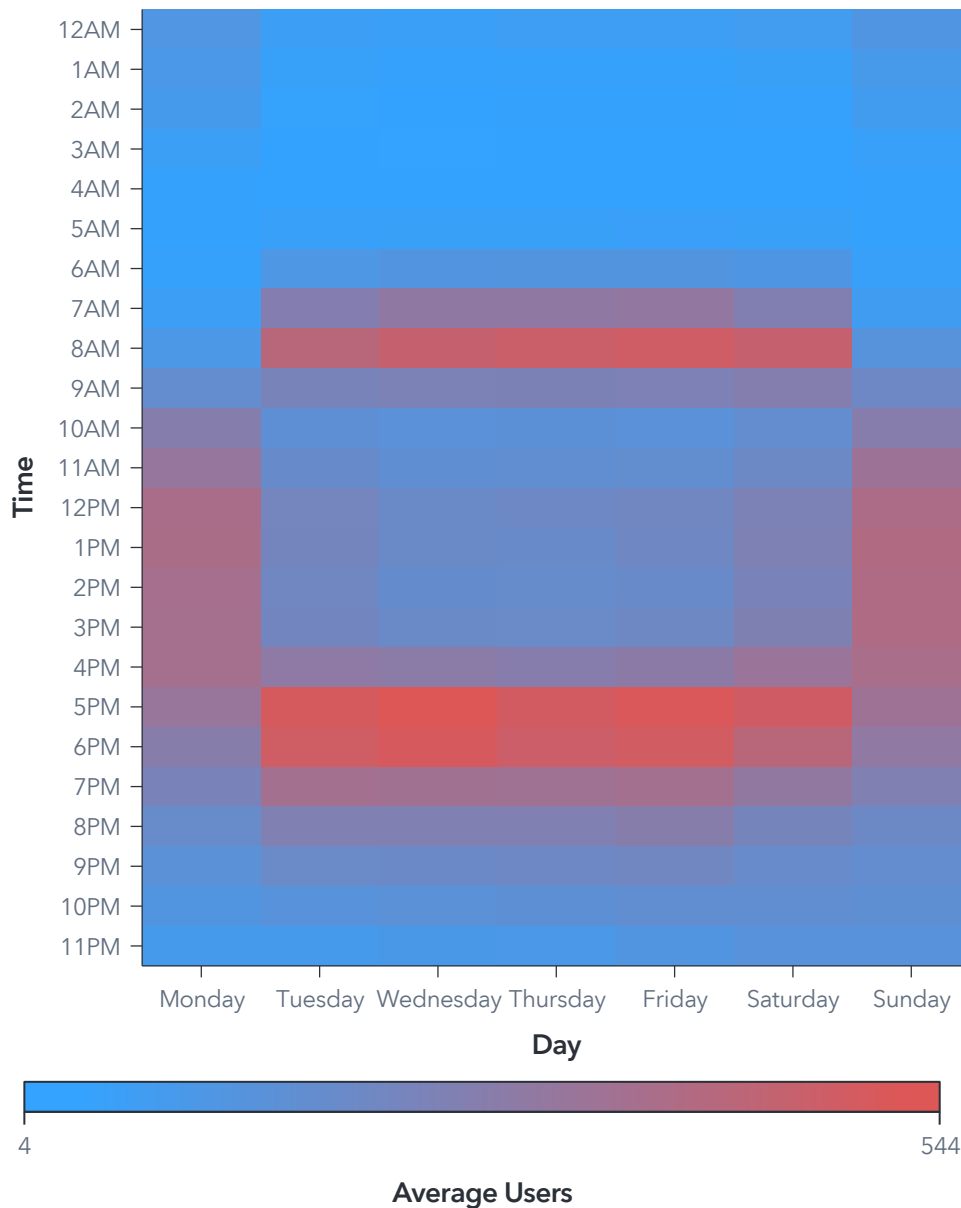


#### Insight 15:

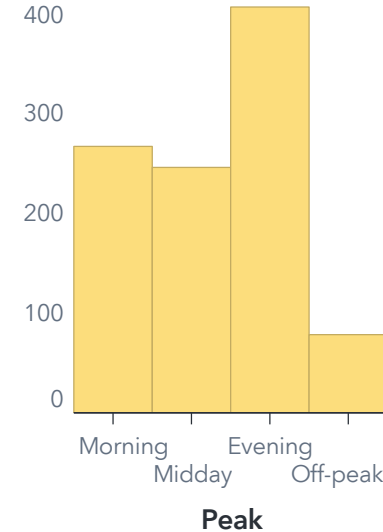
Across the 24 matched hourly slots, **registered > casual** on both day types. On weekdays the mean hourly difference (registered\_avg – casual\_avg) is **101.8** users/hr,  $t(23)=7.46$ ,  $p<.0001$ ; on weekends the difference is **125.9** users/hr,  $t(23)=5.70$ ,  $p<.0001$ . These results confirm that the registered segment leads at virtually every hour, even as the casual share increases on weekends and reshapes the diurnal profile.

## Peak Patterns – Day & Time

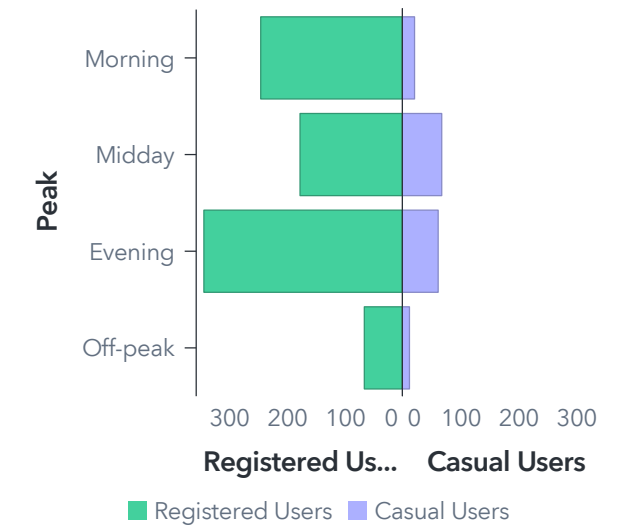
Average Users by Day & Time



Average Users by Peak



Registered Users, Casual Users by Peak



### Insight 16:

Average usage is minimal from midnight to 5 AM, rises sharply after 6-7 AM, and forms two recurrent bands: a morning commute band (about 7-10 AM, strongest mid-week) and a broader evening band (about 4-8 PM, prominent Monday to Friday). Weekends shift later in the day, with a wider mid-afternoon to early-evening band and a softer morning. Sunday shows a visible taper after early evening.

### Insight 17:

**Peak-bucket analysis quantifies the bands:** Evening averages **360.9** users/hr, Midday **244.6**, Morning **243.2**, Off-peak **86.1**. Differences are significant on classical ANOVA ( $F \approx 1806$ ,  $p < .0001$ ) and Welch ( $F \approx 1699$ ,  $p < .0001$ ) with  $R^2 \approx 0.332$ ; Evening exceeds all other buckets, while Morning and Midday are not different. Window-matched tests show **AM commute** higher on weekdays (**weekday – weekend = 62.5 users/hr**,  $t(3) = 5.16$ ,  $p = 0.014$ ), whereas **PM** shows no clear difference (**weekday – weekend = 32.4 users/hr**,  $t(3) = 0.86$ ,  $p = 0.452$ ). These results align with the heatmap: concentrated weekday morning and evening bands, later and broader weekend activity.

## Key Insights

### 1. Weather sets the level:

Average users follow a clear ordering: **Good ≈205, Mist ≈179, Adverse ≈119** users/hr. Stakeholders can read weather as a first-pass demand index.

### 2. Humidity is the strongest suppressor:

Usage begins to decline once humidity exceeds ~40% RH and drops markedly beyond ~60% RH—averaging ≈241 users/hr at ≤60% versus ≈146 users/hr at >60%. Expect materially lower throughput whenever RH is above ~60%.

### 3. Comfort band for temperature:

Demand increases with temperature and **peaks around 30-33 °C**, then softens at the highest temperatures.

### 4. Windspeed is a secondary shaper:

Calm conditions are lower; a broad mid-range plateau from roughly **10-30** holds usage steady. Treat windspeed as a fine-tuning variable, not a primary lever.

### 5. Demand concentrates:

Two repeatable windows dominate capacity: **Morning 07:00-10:00 ≈243 users/hr** and **Evening 16:00-20:00 ≈361 users/hr** (vs **Off-peak ≈86**). Weekday mornings run higher than weekend mornings; evenings are similar.

### 6. Weekend vs weekday level:

Pooled mean is **Weekend ≈197** vs **Weekday ≈189** users/hr (small but consistent uplift). Peaks are higher on weekdays, reflecting commute surges.

### 7. Time Variance:

Time-of-day windows account for the largest share of hour-to-hour variation, followed by humidity; weather category and windspeed explain smaller shares. For planning, prioritize **timing and humidity**.

## Actionable Recommendations

### 1. Hourly completeness and canonical time

**Action:** Guarantee 24 records per day (00-23) with `datetime_local` and `utc_offset`; mark gap-filled hours (`is_missing_filled`).

**Why it matters:** Stabilizes all mean KPIs and commute-window metrics; prevents hidden bias from missing overnight or storm hours.

### 2. Precipitation intensity and duration (beyond drizzle/rain/mist)

**Action:** Add continuous `precip_mm` (or `rain_rate_mm_hr`) and `precip_duration_min`, keeping the existing `precip` category for readability.

**Why it matters:** Rain magnitude is one of the strongest short-horizon suppressors in bikeshare research; separating mist from heavy rain will materially improve model fit and hour-ahead planning.

### 3. Thermal comfort indices (heat + humidity combined)

**Action:** Derive a thermal-comfort index such as **heat index** or **humidex** alongside raw temperature and humidity.

**Why it matters:** Comfort indices summarize joint heat-and-moisture effects more faithfully than temperature alone, improving model interpretability, comparability across days and seasons, and the accuracy of demand forecasts and operational decisions.

### 4. Calendar and event context

**Action:** Persist `is_holiday`, `school_term`, and `major_event` flags per city/zone.

**Why it matters:** Explains systematic exceptions (holiday morning softness, event-driven evening surges) and improves forecast accuracy for specific dates.

### 5. Supply and constraint markers

**Action:** Add `bikes_available`, `docks_available` (station systems) or `rebalance_flag` / `service_zone` (free-floating).

**Why it matters:** Distinguishes true demand dips from stockouts or operational constraints, avoiding misattribution to weather.